

Part C

Course title: SC7 Bayes Methods - OPEN BOOK
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version**

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1. (a) [7 marks] Let $\theta \in \Omega$ be a parameter with prior density $\pi(\theta)$. Let $y = (y_1, \dots, y_n)$, $y \in R^n$ be a realisation of data Y with observation model $Y \sim p(\cdot|\theta)$.
 - (i) Explain what it means for $\pi(\theta|y)$ to be *doubly intractable* and the consequence for MCMC-based estimation of θ given $Y = y$.
 - (ii) Let $S(y) = (S_1(y), \dots, S_p(y))$ be a p -component vector of statistics $S : R^n \rightarrow R^p$ computed from the data. For $S, S' \in R^p$ let $d(S, S')$ be a distance function and let $\delta > 0$ be a simulation parameter. Let $\Delta_y(\delta) = \{y' \in R^n : d(S(Y), S(y)) < \delta\}$. Write down a rejection-ABC algorithm simulating $\pi(\theta|Y \in \Delta_y(\delta))$.
 - (iii) Prove that your rejection-ABC algorithm simulates $\pi(\theta|Y \in \Delta_y(\delta))$.
- (b) [9 marks] The spread of a virus is modeled as a contact process on a square lattice with cells $\mathcal{C} = \{(i, j) : i, j \in \mathbb{Z}\}$. Cells are activated (ie, infected) by their neighbors. Let $\mathcal{N}_{(i,j)} = \{(i-1, j), (i+1, j), (i, j-1), (i, j+1)\}$ be the neighbors of cell (i, j) . Let $0 \leq r \leq 1$ be the recovery probability and $0 \leq a \leq 1/4$ be the activation probability. The set of active cells $X^t \subset \mathcal{C}$ at times $t \in (0, 1, \dots, T)$ is simulated as follows.

Simulation of $X \sim \text{contact}(a, r)$: Let $X_0 = \{(0, 0)\}$. For t from 0 to $T-1$

Update Set $X^{t+1} = X^t$. Let $N_{(i,j)}^t = |\mathcal{N}_{(i,j)} \cap X^t|$ give the number of active neighbors of cell (i, j) at time t . Let $E^t = \{(i, j) \in \mathcal{C} \setminus X^t : N_{(i,j)}^t > 0\}$ be the list of exposed cells.

Activate For $(i, j) \in E^t$ set $X^{t+1} \leftarrow X^{t+1} \cup \{(i, j)\}$ with probability $aN_{(i,j)}^t$.

Deactivate For $(i, j) \in X^t$ set $X^{t+1} \leftarrow X^{t+1} \setminus \{(i, j)\}$ with probability r .

Let $X = X^T$ be the set of all cells active at time T .

 - (i) Suppose we observe $X_{obs} \sim \text{contact}(a, r)$ with a, r unknown but given prior $\pi(a, r)$. Show that $\pi(a, r|X_{obs})$ is in general doubly intractable.
 - (ii) Give a rejection-ABC algorithm targeting $\pi(r, a|X \in \Delta_{X_{obs}}(\delta))$. Your answer should define and justify summary-statistics and distance measure.
- (c) [9 marks] If $\pi(\theta)$ is a prior for θ then an inference scheme is a rule $\psi(\theta; \pi, y)$ for updating belief for θ given data y . For example in Bayesian inference $\psi_{Bayes}(\theta; \pi, y) = \pi(\theta|y)$ but in ABC at fixed δ , $\psi_{\Delta, \delta}(\theta; \pi, y) = \pi(\theta|Y \in \Delta_y(\delta))$. For $1 \leq j < n$ let $y_{1:j} = (y_1, \dots, y_j)$ and $y_{j+1:n} = (y_{j+1}, \dots, y_n)$. Suppose the data are conditionally independent, so

$$p(y|\theta) = \prod_{i=1}^n p(y_i|\theta).$$

A belief update is *order-coherent* for conditionally independent data if

$$\psi(\theta; \pi, y) = \psi(\theta; \psi(\theta; \pi, y_{1:j}), y_{j+1:n})$$

for all $j \in \{1, 2, \dots, n-1\}$ (the posterior from the first data set is the prior for the next).

- (i) Show that Bayesian inference is order-coherent.
- (ii) Show that ABC with fixed δ is not in general order coherent. *Hint: take summary statistic $S(y) = y$ and Euclidean distance measure $d(y, y') = \|y - y'\|$ and give a counter-example.*
- (iii) Let $C_y(\delta)$ be the rectangular prism $C_y(\delta) = \{y' \in R^n : |y_i - y'_i| < \delta \ \forall i = 1, \dots, n\}$. Show that inference with $\psi_{C, \delta}(\theta; \pi, y) = \pi(\theta|Y \in C_y(\delta))$ is order-coherent.

2. Let $X_1, X_2, X_3, \dots$ be an infinite exchangeable sequence of binary random variables. For $n = 1, 2, 3, \dots$ the probability mass functions are $\Pr\{X_1 = x_1, \dots, X_n = x_n\} = p(x_1, \dots, x_n)$.

(a) [10 marks]

- (i) What does it mean to say that these variables are exchangeable?
- (ii) State de Finetti's theorem for the sequence $X_1, X_2, X_3, \dots$
- (iii) Let S be a sample space and let $\mathcal{S}$ be a σ -field of sets in S . In the following events A, B etc are subsets of S and members of $\mathcal{S}$.
For events A, B write $A < B$ if A is less likely a priori than B . Let $\sim$ indicate equality in preference.
Write down the first three of the Savage axioms, in the form given by de Groot.
- (iv) Suppose $A \subseteq B$ (as sets). Show from the axioms that $A \leq B$ (in preference). You may quote results from lectures and classes without proof.

(b) [15 marks]

- (i) Consider the sequence of probability mass functions given for $n = 1, 2, 3, \dots$ by

$$p_n(x) = c_n 2^{-k_n(n-k_n)},$$

with $x = (x_1, \dots, x_n)$, $k_n = \sum_{i=1}^n x_i$ and c_n a normalising constant. Suppose that for each $n = 1, 2, 3, \dots$ our prior probability $P(A)$ for the outcome $A = \{(x_1, \dots, x_n)\}$ is $p_n(x_1, \dots, x_n)$. Prove that the associated prior preferences are not coherent.

Hint: $c_3 = 2/7, c_4 = 8/27$.

- (ii) Show that there does not exist an infinite exchangeable sequence of binary random variables $Y_1, Y_2, Y_3, \dots$ satisfying $\Pr\{Y_1 = x_1, \dots, Y_n = x_n\} = p_n(x_1, \dots, x_n)$ for $n \geq 1$.
- (iii) Consider the sequence of probability mass functions given for $n = 1, 2, 3, \dots$ by

$$q_n(x) = \frac{k_n!(n-k_n)!}{(n+1)!},$$

with $x = (x_1, \dots, x_n)$ and $k_n = \sum_{i=1}^n x_i$ as before. Suppose that for each $n = 1, 2, 3, \dots$ our prior probability $Q(A)$ for the event $A = \{(x_1, \dots, x_n)\}$ is $q_n(x_1, \dots, x_n)$. Prove that the associated prior preferences are coherent.

Hint: is there an infinite exchangeable sequence with these marginals?

3. (a) [5 marks] A random probability distribution $G \sim DP(\alpha, H)$ has a Dirichlet Process distribution with parameter $\alpha > 0$ and base distribution H on $\Omega = R^p$.
- (i) Define the Dirichlet Process $DP(\alpha, H)$.
- (ii) Calculate the expectation and variance of $G(A)$, $A \subseteq \Omega$ in terms of α and $H(A)$, and briefly interpret the role α and H play in determining the distribution of G .

$$\text{If } X \sim \text{Beta}(a, b) \text{ then } E(X) = \frac{a}{a+b} \text{ and } \text{var}(X) = \frac{ab}{(a+b)^2(a+b+1)}.$$

- (b) [10 marks] Consider a linear mixed model with a Dirichlet-Normal mixture model for the random effects. For $i = 1, \dots, n$ let $x_i, y_i \in R$ be covariate and response. For $i = 1, \dots, n$ the observation model is

$$y_i = \alpha_i + x_i \beta_i + \epsilon_i, \quad \text{with } \epsilon_i \sim N(0, \sigma^2)$$

independently. Let $y = (y_1, \dots, y_n)$. Let $\theta_i = (\alpha_i, \beta_i)$ for $i = 1, \dots, n$ and $\theta = (\theta_1, \dots, \theta_n)$. The θ -prior is $\theta_i \sim G$ independent for $i = 1, \dots, n$ with $G \sim DP(\alpha, H)$. The distribution $H(\cdot; \psi)$ has density $h(\cdot; \psi)$, parameters ψ and $\alpha > 0$ is fixed. A prior $\pi(\sigma, \psi)$ is given. Let $\theta^* = (\theta_1^*, \dots, \theta_K^*)$ list the distinct values in θ with $\theta_k^* = (\alpha_k^*, \beta_k^*)$, $k = 1, \dots, K$. Let $S = (S_1, \dots, S_K)$ be a partition of $\{1, 2, \dots, n\}$ satisfying $i \in S_k$ if $\theta_i = \theta_k^*$ and let $n_k = |S_k|$. Let $P_\alpha(S)$ be the prior probability for partition S .

- (i) Interpret the generative model for the data.
- (ii) Suggest an explicit form for the density h (depending on a parameter vector ψ which you should give). Your choice is illustrative, there is no need to justify it.
- (iii) Write down the posterior $\pi(\sigma, \psi, \theta^*, S|y)$ in as much detail as you can.

$$\text{You may use without proof that } P_\alpha(S) = \frac{\alpha^K \prod_{k=1}^K (n_k - 1)!}{\prod_{i=1}^n (\alpha + i - 1)}.$$

- (c) [10 marks] Consider a reversible jump MCMC algorithm targeting $\pi(\sigma, \psi, \theta^*, S|y)$.
- (i) Let $P(S'|S)$ give the probability to propose S' given S . Choose $i \sim U\{1, 2, \dots, n\}$. Let $S^{-i} = (S_1^{-i}, \dots, S_{K-i}^{-i})$ be the partition obtained from $S = (S_1, \dots, S_K)$ by removing element i from its set $i \in S_{k_i}$ in S . Here $K-i = K-1$ if we create an empty cluster when we remove i and otherwise $K-i = K$. Choose $k \in \{1, 2, \dots, K-i+1\}$ with probability p_k , $k = 1, \dots, K-i+1$, where

$$(p_1, \dots, p_{K-i}, p_{K-i+1}) \propto (|S_1^{-i}|, \dots, |S_{K-i}^{-i}|, \alpha).$$

Assign i to set S_k^{-i} : if $k \leq K-i$ set $S' = (S_1^{-i}, \dots, S_k^{-i} \cup \{i\}, \dots, S_{K-i}^{-i})$; if $k = K-i+1$ set $S' = (S_1^{-i}, \dots, S_{K-i+1}^{-i})$ with $S_{K-i+1}^{-i} = \{i\}$. Suppose $S' \neq S$. Show that

$$P(S'|S) = n^{-1} \frac{P_\alpha(S')}{P_\alpha(S^{-i})}.$$

- (ii) Let $X_t = (\sigma, \psi, \theta^*, S)$. Simulate $S' \sim P(S'|S)$ as above.
- If $K' = K$ set $\tilde{\theta}^* = \theta^*$.
 - If $K' = K-1$ set $\tilde{\theta}^* = \theta_{-k_i}^*$, the vector with the parameter for cluster $k_i = \{k : i \in S_k\}$ removed.
 - If $K' = K+1$ set $\tilde{\theta}^* = (\theta_1^*, \dots, \theta_K^*, \theta_{K+1}^*)$ with $\theta_{K+1}^* \sim h(\cdot; \psi)$

With probability $\alpha(\tilde{\theta}^*, S'|\theta^*, S)$ set $X_{t+1} = (\sigma, \psi, \tilde{\theta}^*, S')$ else $X_{t+1} = (\sigma, \psi, \theta^*, S)$.

Calculate $\alpha(\tilde{\theta}^*, S'|\theta^*, S)$ for the case $K' = K+1$.